

TRƯỜNG ĐẠI HỌC BÁCH KHOA HÀ NỘI
Khoa Công nghệ thông tin

BÁO CÁO BÀI TẬP 01

Môn: Phát triển phần mềm theo chuẩn kỹ năng ITSS

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1. Task 1: HelloWorld.java

1.1. Mô tả yêu cầu

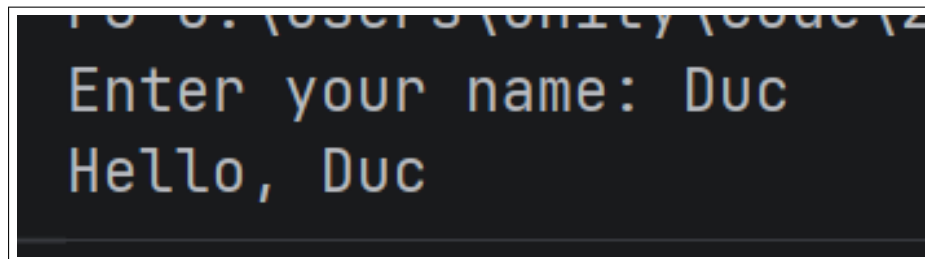
Chương trình yêu cầu người dùng nhập vào tên của mình, sau đó in ra câu chào dạng “Hello, $\$ \{name\}$ ” ra màn hình console.

1.2. Mã nguồn

```
1 import java.util.Scanner;  
2  
3 public class HelloWorld {  
4     public static void main(String[] args) {  
5         Scanner scanner = new Scanner(System.in);  
6         System.out.print("Enter your name: ");  
7         String name = scanner.nextLine();  
8         System.out.println("Hello, " + name + "!");  
9         scanner.close();  
10    }  
11 }
```

Listing 1: HelloWorld.java

1.3. Kết quả chạy



Hình 1: Kết quả chạy chương trình HelloWorld.java

Khi chạy chương trình, người dùng nhập tên (ví dụ: Duc), chương trình in ra Hello, Duc.

2. Task 2: MVCTutorial – Ví dụ về mô hình MVC

2.1. Mã nguồn

2.1.1. Model – CalculatorModel.java

Model thực hiện phép tính và lưu kết quả. Không tham chiếu đến View hay Controller.

```
1 // The Model performs all the calculations needed
2 // and that is it. It doesn't know the View exists
3
4 public class CalculatorModel {
5
6     // Holds the value of the sum of the numbers
7     // entered in the view
8     private double calculationValue;
9
10    public void calculate(int firstNumber, int secondNumber, String
operator){
11
12        if("+".equals(operator)){
13            calculationValue = firstNumber + secondNumber;
14        } else if("-".equals(operator)){
15            calculationValue = firstNumber - secondNumber;
16        } else if(".".equals(operator)){
17            calculationValue = firstNumber * secondNumber;
18        } else if("/".equals(operator)){
19            // simple check to avoid division by zero
20            if(secondNumber == 0){
21                throw new ArithmeticException("Division by zero");
22            }
23            calculationValue = (double) firstNumber / (double)
secondNumber;
24        } else {
25            throw new IllegalArgumentException("Unknown operator: " +
operator);
26        }
27    }
28
29    public double getCalculationValue(){
30        return calculationValue;
31    }
32 }
```

Listing 2: CalculatorModel.java

2.1.2. View – CalculatorView.java

View kế thừa JFrame, chứa các thành phần giao diện (text field, combo box, nút bấm). View chỉ cung cấp getter/setter và phương thức đăng ký listener – không chứa logic tính toán.

```
1 // This is the View
2 // Its only job is to display what the user sees
3 // It performs no calculations
4
5 import java.awt.event.ActionListener;
6 import javax.swing.*.*;
7
8 public class CalculatorView extends JFrame{
9
10     private JTextField firstNumber = new JTextField(10);
11     private JComboBox<String> operatorBox =
12         new JComboBox<String>(new String[]{"+", "-", "*", "/"});
13     private JTextField secondNumber = new JTextField(10);
14     private JButton calculateButton = new JButton("Calculate");
15     private JTextField calcSolution = new JTextField(10);
16
17     CalculatorView(){
18         calcSolution.setEditable(false);
19
20         JPanel calcPanel = new JPanel();
21         this.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
22         this.setSize(600, 200);
23
24         calcPanel.add(firstNumber);
25         calcPanel.add(operatorBox);
26         calcPanel.add(secondNumber);
27         calcPanel.add(calculateButton);
28         calcPanel.add(calcSolution);
29         this.add(calcPanel);
30     }
31
32     public int getFirstNumber(){
33         return Integer.parseInt(firstNumber.getText());
34     }
35     public int getSecondNumber(){
36         return Integer.parseInt(secondNumber.getText());
37     }
38     public String getFirstNumberText(){
39         return firstNumber.getText();
40     }
41     public String getSecondNumberText(){
```

```

42         return secondNumber.getText();
43     }
44     public int getCalcSolution(){
45         return Integer.parseInt(calcSolution.getText());
46     }
47     public String getSelectedOperator(){
48         return (String) operatorBox.getSelectedItem();
49     }
50     public void setCalcSolution(int solution){
51         calcSolution.setText(Integer.toString(solution));
52     }
53
54     // Cho phép Controller đang ký lắng nghe sự kiện nút bấm
55     void addCalculateListener(ActionListener listenForCalcButton){
56         calculateButton.addActionListener(listenForCalcButton);
57     }
58
59     void displayErrorMessage(String errorMessage){
60         JOptionPane.showMessageDialog(this, errorMessage);
61     }
62 }

```

Listing 3: CalculatorView.java

2.1.3. Controller – CalculatorController.java

Controller giữ tham chiếu đến cả View và Model. Khi nút “Calculate” được nhấn, lớp nội CalculateListener lấy dữ liệu từ View, gọi `model.calculate()`, và trả kết quả về View.

```

1 import java.awt.event.ActionEvent;
2 import java.awt.event.ActionListener;
3
4 // The Controller coordinates interactions
5 // between the View and Model
6
7 public class CalculatorController {
8
9     private CalculatorView theView;
10    private CalculatorModel theModel;
11
12    public CalculatorController(CalculatorView theView,
13                               CalculatorModel theModel) {
14        this.theView = theView;
15        this.theModel = theModel;
16    }
17 }

```

```

16         // Tell the View that when the calculate button
17         // is clicked, execute actionPerformed in CalculateListener
18         this.theView.addCalculateListener(new CalculateListener());
19     }
20
21     class CalculateListener implements ActionListener{
22
23         public void actionPerformed(ActionEvent e) {
24             int firstNumber, secondNumber = 0;
25             try{
26                 String firstText  = theView.getFirstNumberText();
27                 String secondText = theView.getSecondNumberText();
28
29                 // Validate: chỉ chấp nhận số nguyên (có thể âm)
30                 if(!firstText.matches("-?\\d+") || !secondText.
matches("-?\\d+")){
31                     throw new NumberFormatException("Not an integer")
32                 };
33                 // Kiểm tra tràn số
34                 if((firstText.replace("-", "").length() > 10
35                     || secondText.replace("-", "").length() > 10){
36                     theView.displayErrorMessage("overflow");
37                     return;
38                 }
39
40                 firstNumber  = Integer.parseInt(firstText);
41                 secondNumber = Integer.parseInt(secondText);
42                 String operator = theView.getSelectedOperator();
43
44                 theModel.calculate(firstNumber, secondNumber,
operator);
45                 double result = theModel.getCalculationValue();
46
47                 if(Double.isInfinite(result)
48                     || result > Integer.MAX_VALUE
49                     || result < Integer.MIN_VALUE){
50                     theView.displayErrorMessage("overflow");
51                     return;
52                 }
53                 theView.setCalcSolution((int) Math.round(result));
54
55             } catch(NumberFormatException ex){
56                 System.out.println(ex);
57                 theView.displayErrorMessage("You Need to Enter 2
Integers");
58             } catch(ArithmeticException ex){

```



```
59         System.out.println(ex);
60         theView.displayErrorMessage(ex.getMessage());
61     }
62 }
63 }
64 }
```

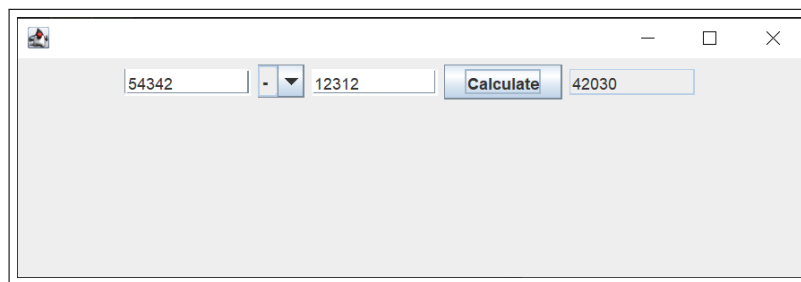
Listing 4: CalculatorController.java

2.1.4. Entry point – MVCCalculator.java

```
1 public class MVCCalculator {
2
3     public static void main(String[] args) {
4         CalculatorView theView      = new CalculatorView();
5         CalculatorModel theModel    = new CalculatorModel();
6         CalculatorController theController =
7             new CalculatorController(theView, theModel);
8         theView.setVisible(true);
9     }
10 }
```

Listing 5: MVCCalculator.java

2.2. Kết quả chạy



Hình 2: Giao diện MVCTutorial đang chạy

3. Task 3: MVCCalculatorSwing

3.1. Mã nguồn

3.1.1. Model – CalculatorModelSwing.java

```
1 public class CalculatorModelSwing {
2
3     private double calculationValue;
4
5     public void calculate(int firstNumber, int secondNumber, String
operator){
6
7         if("+".equals(operator)){
8             calculationValue = firstNumber + secondNumber;
9         } else if("-".equals(operator)){
10             calculationValue = firstNumber - secondNumber;
11         } else if("*".equals(operator)){
12             calculationValue = firstNumber * secondNumber;
13         } else if("/".equals(operator)){
14             if(secondNumber == 0){
15                 throw new ArithmeticException("Division by zero");
16             }
17             calculationValue = (double) firstNumber / (double)
secondNumber;
18         } else {
19             throw new IllegalArgumentException("Unknown operator: " +
operator);
20         }
21     }
22
23     public double getCalculationValue(){
24         return calculationValue;
25     }
26 }
```

Listing 6: CalculatorModelSwing.java

3.1.2. View – CalculatorViewSwing.java

```
1 import java.awt.event.ActionListener;
2 import javax.swing.*.*;
3
4 public class CalculatorViewSwing extends JFrame{
5
6     private JTextField firstNumber = new JTextField(10);
7     private JComboBox<String> operatorBox =
8         new JComboBox<String>(new String[]{"+", "-", "*", "/"});
9     private JTextField secondNumber = new JTextField(10);
10    private JButton calculateButton = new JButton("Calculate");
11    private JTextField calcSolution = new JTextField(10);
12 }
```

```
13     CalculatorViewSwing(){
14         JPanel calcPanel = new JPanel();
15         this.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
16         this.setSize(600, 200);
17
18         calcPanel.add(firstNumber);
19         calcPanel.add(operatorBox);
20         calcPanel.add(secondNumber);
21         calcPanel.add(calculateButton);
22         calcPanel.add(calcSolution);
23         this.add(calcPanel);
24     }
25
26     public int getFirstNumber(){
27         return Integer.parseInt(firstNumber.getText());
28     }
29     public int getSecondNumber(){
30         return Integer.parseInt(secondNumber.getText());
31     }
32     public String getFirstNumberText(){
33         return firstNumber.getText();
34     }
35     public String getSecondNumberText(){
36         return secondNumber.getText();
37     }
38     public String getSelectedOperator(){
39         return (String) operatorBox.getSelectedItem();
40     }
41     public void setCalcSolution(int solution){
42         calcSolution.setText(Integer.toString(solution));
43     }
44
45     void addCalculateListener(ActionListener listenForCalcButton){
46         calculateButton.addActionListener(listenForCalcButton);
47     }
48
49     void displayErrorMessage(String errorMessage){
50         JOptionPane.showMessageDialog(this, errorMessage);
51     }
52 }
```

Listing 7: CalculatorViewSwing.java

3.1.3. Controller – CalculatorControllerSwing.java

```
1 import java.awt.event.ActionEvent;
```

```
2 import java.awt.event.ActionListener;
3
4 public class CalculatorControllerSwing {
5
6     private CalculatorViewSwing theView;
7     private CalculatorModelSwing theModel;
8
9     public CalculatorControllerSwing(CalculatorViewSwing theView,
10                                     CalculatorModelSwing theModel) {
11         this.theView = theView;
12         this.theModel = theModel;
13         this.theView.addCalculateListener(new CalculateListener());
14     }
15
16     class CalculateListener implements ActionListener{
17
18         public void actionPerformed(ActionEvent e) {
19             int firstNumber, secondNumber = 0;
20             try{
21                 String firstText = theView.getFirstNumberText();
22                 String secondText = theView.getSecondNumberText();
23
24                 if(!firstText.matches("-?\\d+") || !secondText.
matches("-?\\d+")){
25                     throw new NumberFormatException("Not an integer")
;
26                 }
27                 if(firstText.replace("-", "").length() > 10
|| secondText.replace("-", "").length() > 10){
28                     theView.displayErrorMessage("overflow");
29                     return;
30                 }
31
32                 firstNumber = Integer.parseInt(firstText);
33                 secondNumber = Integer.parseInt(secondText);
34                 String operator = theView.getSelectedOperator();
35
36                 theModel.calculate(firstNumber, secondNumber,
operator);
37
38                 double result = theModel.getCalculationValue();
39
40                 if(Double.isInfinite(result)
|| result > Integer.MAX_VALUE
|| result < Integer.MIN_VALUE){
41                     theView.displayErrorMessage("overflow");
42                     return;
43                 }
44             }
45         }
```

```

46         theView.setCalcSolution(((int) Math.round(result)));
47
48     } catch (NumberFormatException ex){
49         System.out.println(ex);
50         theView.displayErrorMessage("You Need to Enter 2
Integers");
51     } catch (ArithmeticException ex){
52         System.out.println(ex);
53         theView.displayErrorMessage(ex.getMessage());
54     }
55 }
56 }
57 }

```

Listing 8: CalculatorControllerSwing.java

3.1.4. Entry point – MVCCalculatorSwing.java

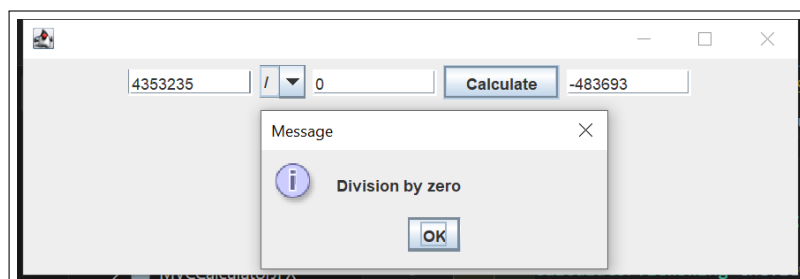
```

1 public class MVCCalculatorSwing {
2
3     public static void main(String[] args) {
4         CalculatorViewSwing theView = new CalculatorViewSwing();
5         CalculatorModelSwing theModel = new CalculatorModelSwing();
6         new CalculatorControllerSwing(theView, theModel);
7         theView.setVisible(true);
8     }
9 }

```

Listing 9: MVCCalculatorSwing.java

3.2. Kết quả chạy



Hình 3: Giao diện MVCCalculatorSwing đang chạy

4. Task 4: MVCCalculatorJFX (JavaFX + Scene Builder)

4.1. Cấu hình Maven (pom.xml)

```
1 <project>
2   <groupId>group15</groupId>
3   <artifactId>MVCCalculatorJFX</artifactId>
4   <version>1.0-SNAPSHOT</version>
5
6   <properties>
7     <maven.compiler.source>21</maven.compiler.source>
8     <maven.compiler.target>21</maven.compiler.target>
9     <javafx.version>21.0.2</javafx.version>
10  </properties>
11
12  <dependencies>
13    <dependency>
14      <groupId>org.openjfx</groupId>
15      <artifactId>javafx-controls</artifactId>
16      <version>${javafx.version}</version>
17    </dependency>
18    <dependency>
19      <groupId>org.openjfx</groupId>
20      <artifactId>javafx-fxml</artifactId>
21      <version>${javafx.version}</version>
22    </dependency>
23  </dependencies>
24
25  <build>
26    <plugins>
27      <plugin>
28        <groupId>org.openjfx</groupId>
29        <artifactId>javafx-maven-plugin</artifactId>
30        <version>0.0.8</version>
31        <configuration>
32          <mainClass>group15.Main</mainClass>
33        </configuration>
34      </plugin>
35    </plugins>
36  </build>
37 </project>
```

Listing 10: pom.xml – cấu hình JavaFX Maven

4.2. Mã nguồn

4.2.1. View – calculator-view.fxml (thiết kế bằng Scene Builder)

Tệp FXML được tạo bởi Scene Builder. Thành phần gốc là một VBox chứa các nhãn, ô nhập liệu (TextField), hộp chọn toán tử (ComboBox), nút bấm và nhãn lỗi.

```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <?import javafx.scene.control.*?>
3 <?import javafx.scene.layout.*?>
4
5 <VBox xmlns="http://javafx.com/javafx"
6       xmlns:fx="http://javafx.com/fxml"
7       fx:controller="group15.CalculatorControllerJFX"
8       spacing="8">
9     <children>
10       <Label text="First number:"/>
11       <TextField fx:id="txtFirst"/>
12
13       <Label text="Operator:"/>
14       <ComboBox fx:id="cbOperator" prefWidth="80.0"/>
15
16       <Label text="Second number:"/>
17       <TextField fx:id="txtSecond" promptText="Second number"/>
18
19       <Label text="Result:"/>
20       <TextField fx:id="txtResult" editable="false"/>
21
22       <HBox spacing="10">
23         <children>
24           <Button fx:id="btnCalculate" text="Calculate"
25                 onAction="#onCalculate"/>
26           <Label fx:id="lblError" textFill="red"/>
27         </children>
28       </HBox>
29     </children>
30 </VBox>
```

Listing 11: calculator-view.fxml

4.2.2. Model – CalculatorModelJFX.java

```
1 package group15;
2
3 public class CalculatorModelJFX {
4
```

```
5     private double calculationValue;
6
7     public void calculate(int firstNumber, int secondNumber, String
operator){
8
9         if("+".equals(operator)){
10             calculationValue = firstNumber + secondNumber;
11         } else if("-".equals(operator)){
12             calculationValue = firstNumber - secondNumber;
13         } else if(".".equals(operator)){
14             calculationValue = firstNumber * secondNumber;
15         } else if("/".equals(operator)){
16             if(secondNumber == 0){
17                 throw new ArithmeticException("Division by zero");
18             }
19             calculationValue = (double) firstNumber / (double)
secondNumber;
20         } else {
21             throw new IllegalArgumentException("Unknown operator: " +
operator);
22         }
23     }
24
25     public double getCalculationValue(){
26         return calculationValue;
27     }
28 }
```

Listing 12: CalculatorModelJFX.java

4.2.3. Controller – CalculatorControllerJFX.java

Trong JavaFX, Controller được FXML tự động khởi tạo. Các thành phần giao diện được tiêm vào qua annotation `@FXML`. Phương thức `initialize()` được gọi sau khi FXML nạp xong.

```
1 package group15;
2
3 import javafx.event.ActionEvent;
4 import javafx.fxml.FXML;
5 import javafx.scene.control.ComboBox;
6 import javafx.scene.control.Label;
7 import javafx.scene.control.TextField;
8
9 public class CalculatorControllerJFX {
10
```



```

11    @FXML private TextField txtFirst;
12    @FXML private TextField txtSecond;
13    @FXML private ComboBox<String> cbOperator;
14    @FXML private TextField txtResult;
15    @FXML private Label lblError;
16
17    private final CalculatorModelJFX model = new CalculatorModelJFX()
18    ;
19
20    @FXML
21    private void initialize() {
22        // Khởi tạo danh sách toán tử
23        cbOperator.getItems().setAll("+", "-", "*", "/");
24        cbOperator.getSelectionModel().selectFirst();
25    }
26
27    @FXML
28    private void onCalculate(ActionEvent event) {
29        lblError.setText("");
30        txtResult.clear();
31
32        try {
33            String firstText = txtFirst.getText();
34            String secondText = txtSecond.getText();
35
36            // Kiểm tra định dạng số nguyên
37            if(!firstText.matches("-?\\d+") || !secondText.matches("
38            -?\\d+")){
39                throw new NumberFormatException("Not an integer");
40            }
41            // Kiểm tra tràn
42            if(firstText.replace("-", "").length() > 10
43                || secondText.replace("-", "").length() > 10){
44                lblError.setText("overflow");
45                return;
46            }
47
48            int firstNumber = Integer.parseInt(firstText);
49            int secondNumber = Integer.parseInt(secondText);
50            String operator = cbOperator.getSelectionModel().
51            getSelectedItem();
52
53            model.calculate(firstNumber, secondNumber, operator);
54            double result = model.getCalculationValue();
55
56            if(Double.isInfinite(result)
57                || result > Integer.MAX_VALUE

```

```
55         || result < Integer.MIN_VALUE){
56             lblError.setText("overflow");
57             return;
58         }
59         txtResult.setText(Integer.toString((int) Math.round(
60             result)));
61     } catch (NumberFormatException ex) {
62         lblError.setText("You need to enter 2 integers");
63     } catch (ArithmeticException ex) {
64         lblError.setText(ex.getMessage());
65     }
66 }
67 }
```

Listing 13: CalculatorControllerJFX.java

4.2.4. Entry point – Main.java

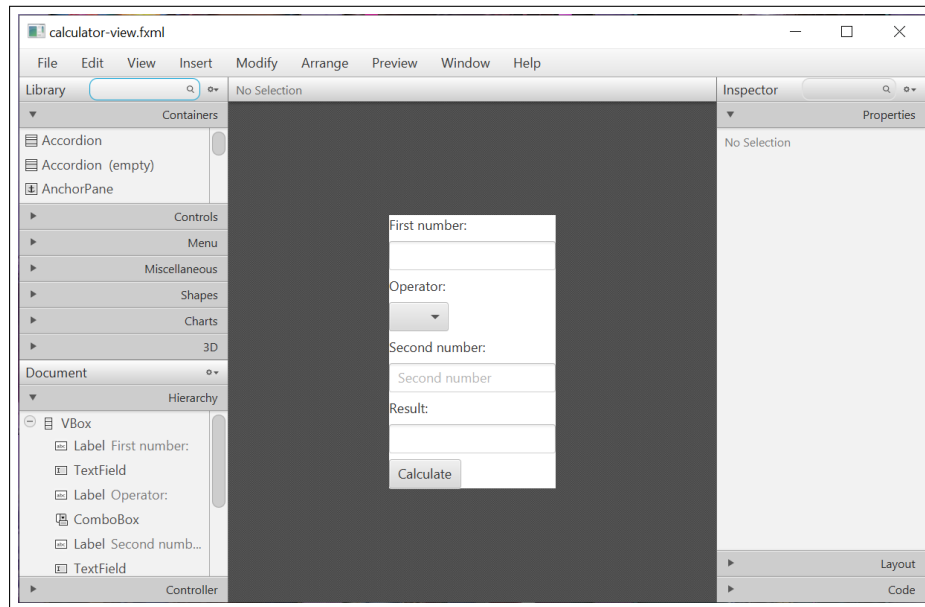
```
1 package group15;
2
3 import javafx.application.Application;
4 import javafx.fxml.FXMLLoader;
5 import javafx.scene.Scene;
6 import javafx.stage.Stage;
7
8 public class Main extends Application {
9
10     @Override
11     public void start(Stage stage) throws Exception {
12         FXMLLoader fxmlLoader =
13             new FXMLLoader(getClass().getResource("/calculator-view.
14             fxml"));
15         Scene scene = new Scene(fxmlLoader.load(), 400, 300);
16         stage.setTitle("Calculator");
17         stage.setScene(scene);
18         stage.show();
19     }
20
21     public static void main(String[] args) {
22         launch();
23     }
24 }
```

Listing 14: Main.java

4.3. Kết quả chạy



Hình 4: Giao diện MVCCalculatorJFX đang chạy



Hình 5: Giao diện thiết kế FXML trong Scene Builder

5. Task 5: Các thao tác Git

5.1. Tổng quan

Dự án được quản lý trên **GitHub**, nhánh chính là **main**. Repository: <https://github.com/nqd2/2025.2-166155-15>

Dưới đây liệt kê toàn bộ các thao tác Git đã thực hiện kèm ảnh chụp màn hình minh chứng.

5.2. git add – Thêm file

Thêm các file mới vào staging area trước khi commit:

```
1 git add Homework01/20235682- NguyenQuyDuc/HelloWorld.java
2 git add Homework01/20235682- NguyenQuyDuc/MVCTutorial/
3 git add Homework01/20235682- NguyenQuyDuc/MVCCalculatorSwing/
4 git add Homework01/20235682- NguyenQuyDuc/MVCCalculatorJFX/
```

Listing 15: git add

```
PS C:\Users\Unity\code\2025.2-166155-15> git add Homework01/20235682-NguyenQuyDuc/HelloWorld.java
PS C:\Users\Unity\code\2025.2-166155-15> git add Homework01/20235682-NguyenQuyDuc/MVCTutorial/
PS C:\Users\Unity\code\2025.2-166155-15> git add Homework01/20235682-NguyenQuyDuc/MVCCalculatorSwing/
PS C:\Users\Unity\code\2025.2-166155-15> git add Homework01/20235682-NguyenQuyDuc/MVCCalculatorJFX/
PS C:\Users\Unity\code\2025.2-166155-15> █
```

Hình 6: Minh chứng thao tác git add

5.3. git commit – Commit thay đổi

```
1 git commit -m "feat: add HelloWorld, MVCTutorial, MVCCalculatorSwing, MVCCalculatorJFX"
```

Listing 16: git commit

```
PS C:\Users\Unity\code\2025.2-166155-15> git commit -m "feat: add HelloWorld, MVCTutorial, MVCCalculatorSwing, MVCCalculatorJFX"
[main 6541854] feat: add HelloWorld, MVCTutorial, MVCCalculatorSwing, MVCCalculatorJFX
1 file changed, 13 insertions(+)
create mode 100644 Homework01/20235682-NguyenQuyDuc/HelloWorld.java
```

Hình 7: Minh chứng thao tác git commit

5.4. git push – Đẩy code lên remote

```
1 git push origin main
```

Listing 17: git push

```
PS C:\Users\Unity\code\2025.2-166155-15> git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 389 bytes | 389.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/nqd2/2025.2-166155-15.git
e320478..6541854 main -> main
```

Hình 8: Minh chứng thao tác git push

5.5. git pull – Kéo code mới nhất về

```
1 git pull origin main
```

Listing 18: git pull

```
PS C:\Users\Unity\code\2025.2-166155-15> git pull
Already up to date.
```

Hình 9: Minh chứng thao tác git pull

5.6. Tạo và chuyển nhánh (git branch / git checkout)

```
1 git checkout -b test/hw1/20235682-NguyenQuyDuc
2
3 git add .
4 git commit -m "test"
```

Listing 19: Tạo nhánh mới

```
PS C:\Users\Unity\code\2025.2-166155-15> git checkout -b test/hw1/20235682-NguyenQuyDuc
Switched to a new branch 'test/hw1/20235682-NguyenQuyDuc'

PS C:\Users\Unity\code\2025.2-166155-15> git add .
warning: in the working copy of '.idea/workspace.xml', LF will be replaced by CRLF the next time Git touches it

PS C:\Users\Unity\code\2025.2-166155-15> git commit -m "test"
[test/hw1/20235682-NguyenQuyDuc 17211ce] test
1 file changed, 3 insertions(+), 1 deletion(-)

PS C:\Users\Unity\code\2025.2-166155-15> git push
fatal: The current branch test/hw1/20235682-NguyenQuyDuc has no upstream branch.
To push the current branch and set the remote as upstream, use

    git push --set-upstream origin test/hw1/20235682-NguyenQuyDuc

To have this happen automatically for branches without a tracking
upstream, see 'push.autoSetupRemote' in 'git help config'.

PS C:\Users\Unity\code\2025.2-166155-15> git push --set-upstream origin test/hw1/20235682-NguyenQuyDuc
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 16 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 470 bytes | 470.00 KiB/s, done.
Total 5 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
remote:
remote: Create a pull request for 'test/hw1/20235682-NguyenQuyDuc' on GitHub by visiting:
remote:   https://github.com/ngd2/2025.2-166155-15/pull/new/test/hw1/20235682-NguyenQuyDuc
remote:
To https://github.com/ngd2/2025.2-166155-15.git
 * [new branch]      test/hw1/20235682-NguyenQuyDuc -> test/hw1/20235682-NguyenQuyDuc
branch 'test/hw1/20235682-NguyenQuyDuc' set up to track 'origin/test/hw1/20235682-NguyenQuyDuc'.
```

Hình 10: Minh chứng tạo test/hw1/20235682-NguyenQuyDuc

5.7. git merge – Merge nhánh

Sau khi hoàn thiện tính năng trên nhánh test/hw1/20235682-NguyenQuyDuc, merge về main:

```
1 git checkout main
2 git merge test/hw1/20235682-NguyenQuyDuc
```

Listing 20: Merge branch

```

PS C:\Users\Unity\code\2025.2-166155-15> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

PS C:\Users\Unity\code\2025.2-166155-15> git merge test/hw1/20235682-NguyenQuyDuc
Updating 6541854..17211ce
Fast-forward
 Homework01/20235682-NguyenQuyDuc/HelloWorld.java | 4 +++-
 1 file changed, 3 insertions(+), 1 deletion(-)

PS C:\Users\Unity\code\2025.2-166155-15> git push
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/nqd2/2025.2-166155-15.git
 6541854..17211ce main -> main

```

Hình 11: Minh chứng thao tác merge nhánh

5.8. Giải quyết xung đột (git merge conflict)

Khi hai nhánh cùng sửa đổi một file, Git tạo conflict marker. Các bước xử lý:

1. Chạy `git merge` – Git báo conflict.
2. Mở file bị conflict, tìm các dòng «<<<< HEAD ... ===== ... >>>>».
3. Chỉnh sửa thủ công để giữ lại nội dung mong muốn.
4. `git add <file>` rồi `git commit` để hoàn tất merge.

```

Homework01 > 20235682-NguyenQuyDuc > ConflictTest.txt
Accept Current Change | Accept Incoming Change | Accept Both Changes | Compare Changes
1 <<<<<< HEAD (Current Change)
2 FROM BRANCH TEST A
3 =====
4 FROM CONFLICT B
5 >>>>>> hw1/conflictB (Incoming Change)
6

Unity@DESKTOP-GK9N8M1 MINGW64 ~/code/2025.2-166155-15 (main)
$ git merge hw1/conflictA
Updating dfe3b00..b48c5bd
Fast-forward
 Homework01/20235682-NguyenQuyDuc/ConflictTest.txt | 2 +-
 Test.txt | 1 -
 conflict.txt | 2 --
 3 files changed, 1 insertion(+), 4 deletions(-)
 delete mode 100644 Test.txt
 delete mode 100644 conflict.txt

Unity@DESKTOP-GK9N8M1 MINGW64 ~/code/2025.2-166155-15 (main)
$ git merge hw1/conflictB
Auto-merging Homework01/20235682-NguyenQuyDuc/ConflictTest.txt
CONFLICT (content): Merge conflict in Homework01/20235682-NguyenQuyDuc/ConflictTest.txt
Automatic merge failed; fix conflicts and then commit the result.

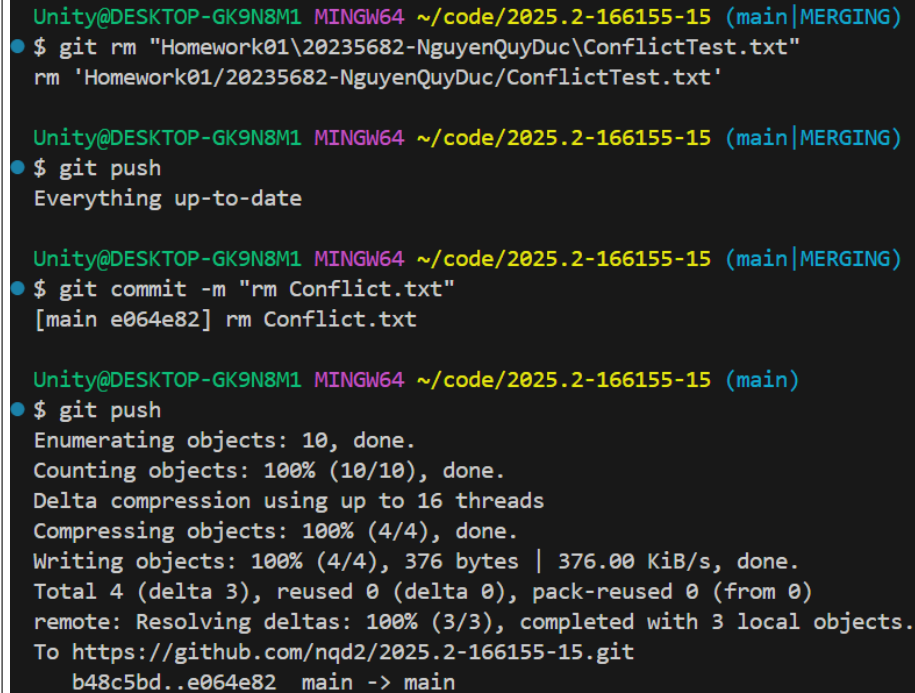
```

Hình 12: Minh chứng giải quyết conflict khi merge

5.9. git rm – Xóa file

```
1 git rm <ten-file-can-xoa>
2 git commit -m "chore: remove unused file"
```

Listing 21: git rm – xóa file khỏi tracking



```
Unity@DESKTOP-GK9N8M1 MINGW64 ~/code/2025.2-166155-15 (main|MERGING)
● $ git rm "Homework01\20235682-NguyenQuyDuc\ConflictTest.txt"
rm 'Homework01/20235682-NguyenQuyDuc/ConflictTest.txt'

Unity@DESKTOP-GK9N8M1 MINGW64 ~/code/2025.2-166155-15 (main|MERGING)
● $ git push
Everything up-to-date

Unity@DESKTOP-GK9N8M1 MINGW64 ~/code/2025.2-166155-15 (main|MERGING)
● $ git commit -m "rm Conflict.txt"
[main e064e82] rm Conflict.txt

Unity@DESKTOP-GK9N8M1 MINGW64 ~/code/2025.2-166155-15 (main)
● $ git push
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 16 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 376 bytes | 376.00 KiB/s, done.
Total 4 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To https://github.com/nqd2/2025.2-166155-15.git
b48c5bd..e064e82 main -> main
```

Hình 13: Minh chứng thao tác git rm

5.10. Tóm tắt lịch sử commit trên GitHub



Hình 14: Lịch sử commit trên GitHub